

Executive Business Report

End-to-End Sales Forecasting & Demand Intelligence System

Executive Summary

This project develops an end-to-end sales forecasting and demand intelligence system for retail businesses. Historical sales data was analyzed to identify trends, forecast future demand, detect anomalies, and segment products based on demand behavior. Three forecasting models (SARIMA, Prophet, and XGBoost) were implemented and compared using MAE, RMSE, and MAPE. An interactive Streamlit dashboard was developed to present insights for business users.

Key Findings

- Historical sales exhibit trend and seasonal patterns.
- Forecasting models were evaluated using MAE, RMSE, and MAPE.
- Anomaly detection identified unusual sales spikes and drops.
- Product demand segmentation supports inventory optimization.

3-Month Sales Forecast

The selected forecasting model estimates sales for the next three months, helping improve inventory planning and reduce stock shortages or overstocking. Confidence intervals indicate the uncertainty of predictions.

Top 3 Anomalies

1. High sales spike due to seasonal promotions.
2. Sudden sales decline possibly caused by supply-chain or stock issues.
3. Unexpected demand increase linked to marketing campaigns or popular products.

Demand Segmentation & Stocking Strategy

High-demand products should maintain higher inventory levels. Stable-demand products should follow regular replenishment schedules. Low-demand and volatile products should be monitored carefully to minimize inventory costs.

Business Recommendations

1. Adopt the best-performing forecasting model for demand planning.
2. Monitor anomaly reports regularly to identify unusual business events.
3. Use demand clusters to optimize inventory allocation and replenishment.

Project Limitation

Forecast accuracy depends on historical data quality and may be affected by unexpected market changes, promotions, or external events.

Conclusion

This project demonstrates how forecasting, anomaly detection, clustering, and interactive dashboards can support data-driven business decisions and improve inventory management.